

Robotic Microscope Signal-Level Schematics
Detailed Pin-Level Connections

TEENSY 4.1 PINOUT

Teensy 4.1 Pin Assignments			
Pin	Function	Direction	Signal
0	UART RX	←	Serial1 RX (from RPi)
1	UART TX	→	Serial1 TX (to RPi)
2	X_STEP	→	Step pulse (X motor)
3	X_DIR	→	Direction (X motor)
4	X_MIN	←	Limit switch (pullup)
5	X_MAX	←	Limit switch (pullup)
6	Y_STEP	→	Step pulse (Y motor)
7	Y_DIR	→	Direction (Y motor)
8	Y_MIN	←	Limit switch (pullup)
9	Y_MAX	←	Limit switch (pullup)
10	Z_STEP	→	Step pulse (Z motor)
11	Z_DIR	→	Direction (Z motor)
12	Z_MIN	←	Limit switch (pullup)
13	LED	-	Onboard (unused)
14	Z_MAX	←	Limit switch (pullup)
15	F_STEP	→	Step pulse (F motor)
16	F_DIR	→	Direction (F motor)
18	F_MAX	←	Limit switch (pullup)
22	F_MIN	←	Limit switch (pullup)
33	ENA_ALL	→	Enable (all TB6600s)
GND	Ground	-	Multiple pins available
5V	Power In	-	From buck converter

RASPBERRY PI 4 GPIO

RPi4 GPIO Connections		
GPIO	Function	Connection
10	SPI0 MOSI	LED Ring (via level shifter)
14	UART TX	Teensy Pin 0 (RX)
15	UART RX	Teensy Pin 1 (TX)
GND	Ground	Common ground bus
5V	Power In	From buck converter
Camera: CSI ribbon → HDMI adapter		

UART CONNECTION (RPi ↔ Teensy)

3-Wire UART (115200 baud)		
RPi4 GPIO	Signal	Teensy Pin
GPIO 14 (TX)	→ Data →	Pin 0 (RX)
GPIO 15 (RX)	← Data ←	Pin 1 (TX)
GND	— Common —	GND
Parameters Baud: 115200 Data: 8 bits Parity: None Stop: 1 Flow: None		
⚠ Both use 3.3V logic No level shifting needed		

LED RING CONTROL (with Level Shifter)

NeoPixel LED Ring Circuit		
SN74AHCT125N Level Shifter (14-pin DIP)		
Pin	Function	Connection
14	VCC	+5V (from buck converter)
7	GND	Ground (common bus)
1	1A (Input)	RPi GPIO10 (3.3V MOSI)
2	1Y (Output)	LED Ring DIN (5V)
1	1OE (Enable)	GND (active LOW = ON)
3,5,9,11	Unused Inputs	Tie to GND
WS2812B LED Ring (24 LEDs)		
Pin	Function	Connection
DIN	Data In	From 74AHCT125 pin 2 (1Y)
5V	Power	+5V rail (buck converter)
GND	Ground	Ground bus
Notes: - WS2812B requires 5V logic - RPi GPIO is 3.3V 74AHCT125 shifts 3.3V → 5V - Current @ 40%: ~500mA - Current @ 70% max: ~900mA		

TB6600 STEPPER DRIVER CONNECTIONS

TB6600 Driver Wiring (4× drivers identical)					
Axis	PUL+	DIR+	ENA+	Motor	Power
X	Pin 2	Pin 3	Pin 33	4-wire	24V/GND
Y	Pin 6	Pin 7	Pin 33	4-wire	24V/GND
Z	Pin 10	Pin 11	Pin 33	4-wire	24V/GND
F	Pin 15	Pin 16	Pin 33	4-wire	24V/GND
Control Signal Details					
Terminal	+ Connection		- Connection (GND)		
PUL	Teensy STEP pin (3.3V)		Common Ground		
DIR	Teensy DIR pin (3.3V)		Common Ground		
ENA	Pin 33 (shared)		Common Ground		
Note: All (-) terminals connect to common ground bus					
⚠ Enable Logic: Active LOW Pin 33 LOW = Drivers ENABLED Pin 33 HIGH = Drivers DISABLED					
Motor Connections (NEMA17 Bipolar)					
Terminal	Function				
A+/A-	Coil A (typically Red/Green or Black/Red)				
B+/B-	Coil B (typically Blue/White or Yellow/Green)				
Use multimeter to identify coil pairs (continuity test)					
DIP Switch Configuration					
Current Limit: 1A (per TB6600 datasheet) Microstepping: 1/16 (SW5-8: All ON) Calibrated: 400 steps/mm					

LIMIT SWITCH WIRING

Limit Switch Connections (8× switches)			
Axis	Switch	Teensy Pin	Wiring
X	X_MIN	4	Pin — Switch — GND (Normally Open)
	X_MAX	5	
Y	Y_MIN	8	
	Y_MAX	9	
Z	Z_MIN	12	
	Z_MAX	14	
F	F_MIN	22	
	F_MAX	18	
Configuration			
• Type: Mechanical NO (Normally Open) • Pull-up: Internal (INPUT_PULLUP) • Logic: Active LOW (pressed = GND) • Debounce: 50ms in firmware • Interrupt: FALLING edge • Connector: 2-pin JST-SM			

CAMERA CONNECTION

Camera Module	
Interface	CSI (Camera Serial Interface)
Connection	15-pin ribbon cable
Adapter	CSI → HDMI converter
RPi Port	CSI camera port (near HDMI)
Note: Currently prototyping with USB Production uses CSI for latency	

EXTERNAL CONNECTOR SUMMARY

JST-SM Connectors (External)			
Component	Pins	Signals	Qty
Stepper Motors	4	A+, A-, B+, B-	4
Limit Switches	2	Signal, GND	8
LED Ring	3	DIN, 5V, GND	1
eStop (future)	2	NC contact	1
Internal Connections (screw terminals)			
Power Input	2	24V, GND	1
TB6600 Power	2	24V, GND	4
TB6600 Control	6	PUL+/-, DIR+/-, ENA+/-	4
UART (RPi-Teensy)	3	TX, RX, GND	1